Simulation of a Seven-Engine Recoverable Booster with Two-Phase Retropropulsion

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Abstract

This paper presents the simulation study of a 130 t dry-mass booster separating at 70 km altitude with a speed of 6,000 km/h and an upward flight-path angle of 45°. Seven methane–oxygen engines support entry braking and terminal descent. The study combines variable mass, altitude-dependent gravity, atmospheric density, Mach-dependent drag, wind, and state-dependent guidance, and examines initial propellant loads between 90 and 200 t.

The baseline results show entry-burn consumption near 40–44 t, while terminal consumption increases with carried mass. A fine baseline scan reports a lowest successful tested load of 104 t and a disturbance-tested load of 120 t. A four-parameter Powell and grid search shifts entry ignition from 75 to 55 km, reduces the reported nominal threshold to 96 t, and supports a 110 t recommendation in its selected disturbance set. Subsequent area-law and finite-response refinements give a reported threshold of 102 t and a practical recommendation of 110–112 t. At 110 t, the combined refined model consumes 106.6 t and reaches a reported touchdown speed of 2.02 m/s.

Separate crosswind, reaction-control, and jet-interaction analyses characterize effects omitted from the simplest baseline. The findings emphasize the interaction between burn timing, carried mass, throttle floor, and engine-count staging. These results belong to distinct model versions; their empirical thresholds are not global minimum-fuel proofs. Heating, fully coupled attitude/contact dynamics, and the completeness of uncertainty coverage remain limitations.

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1 Introduction

1.1 Problem background and restatement

Recovering a first-stage booster requires a controlled return from high speed to a nearly vertical, low-speed arrival at a ship. Atmospheric drag assists the descent, but the same flow produces pressure loads and wind-induced deviations. Propellant retained at separation increases braking capability while also increasing the mass that must be decelerated. Burn timing and engine selection therefore influence both fuel consumption and terminal control.

The booster has dry mass 130,000 kg, height 65 m, diameter 7.0 m, and seven 1 MN-class liquid-oxygen/methane engines. Separation occurs at 70 km with speed 1,666.667 m/s and flight-path angle +45°. The ship can be positioned downrange as allowed by the competition problem [1]. Initial total propellant is varied from 90 to 200 t, followed by finer scans near the feasibility boundaries of the refined models.



Figure 1. A Falcon 9 booster during a powered droneship landing (CRS-8). Photograph: SpaceX, CC0. The photograph illustrates the recovery context; the modeled seven-engine vehicle is hypothetical.

1.2 Objectives and study organization

The study addresses four questions: how the ascent and return arc develop after upward separation; how entry and terminal burns divide the propellant budget; whether reduced loads remain feasible under the reported perturbations; and how area coupling, engine transients, and engine-count changes modify the load recommendation.

The analysis distinguishes baseline guidance, parameterized guidance, and actuator refinements. Their thresholds and recommendations are compared within their respective assumptions and perturbation sets.

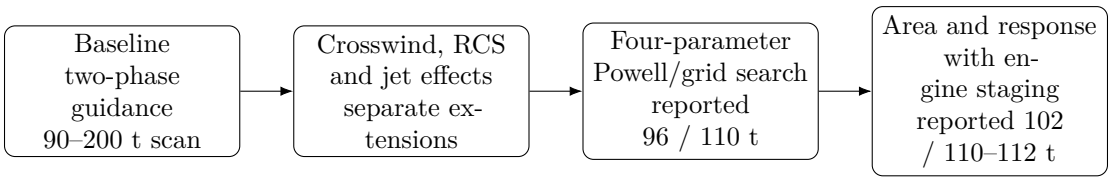


Figure 2. Organization of the simulation study. The successive boxes describe reported study stages, not a claim that every physical extension was coupled into one validated simulator. Nominal thresholds and recommended loads belong to their respective versions.

2 Assumptions, Parameters, and Notation

2.1 Reference assumptions

The baseline motion is described in the vertical plane determined by the separation velocity and Earth center. Gravity varies with altitude, mass decreases as propellant is expelled, and drag is evaluated relative to the prescribed wind. The baseline guidance treats attitude alignment as ideal; later sections introduce separate attitude estimates and a first-order lag.

The initial trajectory is upward, with an apogee near 143 km. The atmospheric representation therefore includes an exponential continuation above 86 km, together with a Mach-dependent drag coefficient.

Table 1. Parameters of the simulation study.

Parameter	Value	Basis / status
Dry mass; dimensions	130 t; 65 m; 7.0 m	Problem data (G)
Separation altitude; speed	70 km; 1666.667 m/s	Problem data (G)
Initial flight-path angle	45°	Scenario (H)
Engine system	7×1 MN; $I_{sp} = 330$ s	Virtual reference (H)
Oxygen/methane ratio	3.4	Scenario (H)
Reference area	38.4845 m^2	Derived from diameter (D)
Mach-dependent drag	12 anchors; maximum $C_D = 1.50$	Simulation drag model (H)
Atmosphere	USSA 1976 layers; exponential above 86 km	Reference plus extension
Earth	$R_E = 6371$ km; $\mu = 3.986 \times 10^{14}$ m^3/s^2	Reference constants
Wind profile	Height-dependent; see Table 3	Scenario (H)
Pressure limit	50 kPa	Screening target (H)
Target touchdown speed	1.5 m/s; baseline test below 3 m/s	Guidance and acceptance choices (H)

G denotes problem data, H a scenario or modeling assumption, and D a value derived from the geometry. The drag representation has 12 anchors and a peak coefficient of 1.50. The complete anchor set and the high-altitude scale height require specification for independent numerical reproduction.

2.2 Notation and angular convention

Table 2. Notation used to express the study model.

Symbol	Meaning
s	Surface range
h	Altitude

Symbol	Meaning (continued)
r	Geocentric radius
R_E	Earth radius
u	Horizontal ground velocity
w	Vertical ground velocity
V_g	Ground speed
V_{rel}	Air-relative speed
m	Total vehicle mass
$m_{p,0}$	Initial total propellant mass
T	Thrust magnitude
D	Drag magnitude
β_h	Thrust angle from the local horizontal in the force equations
χ	Drag-direction angle
γ	Flight-path angle
θ_T	Reported thrust deflection from the upward vertical
ρ	Air density
C_D	Drag coefficient
M	Mach number
q	Dynamic pressure
h_e	Entry ignition height
v_{cut}	Entry cutoff speed
k	Terminal ignition multiplier
V_{td}	Target touchdown speed
I	Moment of inertia
θ	Body rotation angle
M_{RCS}	Reaction-control-system moment
C_T	Thrust coefficient

The thrust angle β_h is measured from the local horizontal in the force-component equations. The deflection from the upward vertical is denoted separately by θ_T. These angular conventions must be kept distinct when interpreting the thrust-direction plots.

3 Physical Model and Two-Phase Guidance

3.1 Geometry, gravity, and variable mass

The model uses state variables (s, h, u, w, m) with

$$r = R_E + h, \qquad \frac{ds}{dt} = \frac{R_E u}{r}, \qquad \frac{dh}{dt} = w. \tag{1}$$

The translational acceleration and mass equations are

$$\frac{du}{dt} = \frac{T\cos\beta_h + D\cos\chi}{m}, \tag{2}$$

$$\frac{dw}{dt} = \frac{T\sin\beta_h + D\sin\chi}{m} - g(h), \tag{3}$$

$$\frac{dm}{dt} = -\frac{T}{I_{sp} g_0}, \qquad g(h) = \frac{\mu}{r^2}. \tag{4}$$

The drag-direction angle χ already points opposite the air-relative velocity, so its components carry the appropriate signs. Earth reference values and the standard atmosphere are associated with the study references [2, 3, 4].

Equations (1)–(4) combine a spherical range conversion with a local Cartesian acceleration form. They omit local velocity-basis curvature terms and must therefore be interpreted as an approximate closure rather than an exact curved-Earth formulation.

The oxygen-to-methane mass ratio is 3.4. All reported fuel amounts below are total propellant, including both liquids; they are not methane-only masses. Initial carried propellant and consumed propellant are distinguished because successful cases can retain a residual load.

3.2 Atmosphere, drag, and wind

The model describes the lower atmosphere using the geopotential conversion and layerwise temperature relation

$$H = \frac{R_a h}{R_a + h}, \qquad T_a = T_b + L_b(H - H_b), \qquad \rho = \frac{p}{R_{air} T_a}. \tag{5}$$

The lower model follows USSA 1976, with an exponential extension above 86 km [2]. An air-relative velocity is formed before computing drag:

$$\boldsymbol{v}_{rel} = \boldsymbol{v} - \boldsymbol{w}_{wind}(h), \qquad D = \tfrac{1}{2}\rho V_{rel}^2 C_D(M_{rel}) A, \qquad M_{rel} = \frac{V_{rel}}{a(h)}. \tag{6}$$

The initial reference area is $A = \pi d^2/4 = 38.4845$ m^2. The later attitude-dependent area law is introduced separately in section 7.

Table 3. Along-track wind profile anchors.

Altitude	Wind (m/s)	Interpretation
0–100 m	8	Sea wind
2 km	15	Boundary layer
11 km	35	Jet-level anchor
20 km	20	Lower stratosphere
Above 20 km	$20e^{-(h_{\mathrm{km}}-20)/15}$	Exponential decay

The wind values in table 3 are prescribed assumptions. Above 20 km, the height in the exponential is measured in kilometers. They are not measured weather at a specified launch location or date. Wind contributes through relative airflow; the terminal guidance instead aims to reduce ground velocity so that the vehicle can approach the ship rather than drift with the air.

3.3 Entry-burn rule

The baseline entry ignition occurs during descent through 75 km. All seven engines operate at a combined nominal thrust of 7 MN, directed opposite the air-relative velocity. The guidance ends that pulse at 55 km. Its main role is to remove roughly 300–350 m/s before the denser atmosphere, rather than completing the entire landing deceleration.

Because this phase follows a high-altitude coast, its propellant use changes relatively little across the study load scan: about 40–44 t. The heavier cases still require a different terminal burn because their thrust-to-mass ratio and braking distance differ.

3.4 Terminal ignition and throttle law

The guidance triggers terminal braking using a stopping-distance condition or a pressure-related speed trigger. With $V_g = \sqrt{u^2 + w^2}$ and $a_{eff} = T_{max}/m - g$, the written rules are

$$h \leq 1.1 \frac{V_g^2 - V_{td}^2}{2a_{eff}}, \tag{7}$$

$$V \geq \sqrt{\frac{2q_{lim}}{\rho C_D}}, \tag{8}$$

$$T = m \left[\frac{V_g^2 - V_{td}^2}{2h} + \frac{g|w|}{V_g} \right] - D. \tag{9}$$

The baseline clips thrust to $[0, 7]$ MN, uses target $V_{td} = 1.5$ m/s, and requests full thrust if $q > 0.95 q_{lim}$ with $q_{lim} = 50$ kPa.

Equation (8) is an empirical control trigger. Since $q = \rho V_{rel}^2/2$, a pure dynamic-pressure speed bound would not include C_D. The trigger implementation requires verification before independent numerical reproduction. A full-throttle request near the limit is a control action, not proof that the subsequent pressure peak remains below the limit.

3.5 Numerical method and model versions

The numerical method is fixed-step fourth-order Runge–Kutta integration with $\Delta t = 0.02$ s and a linearly interpolated landing event. Independent solver execution and runtime benchmarking have not been verified in this analysis.

Four reported study stages are distinguished: baseline two-phase guidance; separate three-dimensional, attitude, and jet investigations; four-parameter Powell/grid guidance search; and the later combined area-law, actuator-response, and engine-staging model. The different success thresholds and recommendations are compared as separate records, not merged into a single mathematical minimum.

4 Baseline Trajectory and Propellant Results

4.1 Trajectory and flight histories

Figure 3 retains the study trajectory curves. Loads have similar early arcs because the unpowered high-altitude motion is dominated by gravity. The approximate overlap should not be generalized to exact mass independence in a finite-drag model. Later differences emerge as the heavier cases begin terminal braking sooner.

The results indicate an apogee near 143 km at about 127 s. The return reaches the entry-ignition height near 245 s. The entry pulse then reduces speed before the longer terminal braking phase. Ground speed, flight-path angle, and altitude histories are shown separately so that vehicle direction is not inferred from an altitude curve alone.

4.2 Load scan and mass allocation

In the baseline records, 90 t and 100 t fail through propellant exhaustion. At 110 t the reported consumption is 42.7 t in entry and 64.5 t in terminal braking. At 200 t those figures become 40.1 t and 120.6 t. Thus additional carried mass does not translate into the same amount of additional residual propellant: it also changes the braking effort and the time spent under gravity.

The model's finer baseline scan reports 103 t arriving at 5.8 m/s and 104 t arriving at 2.48 m/s with about 0.2 t remaining. We refer to 104 t as the lowest reported successful tested

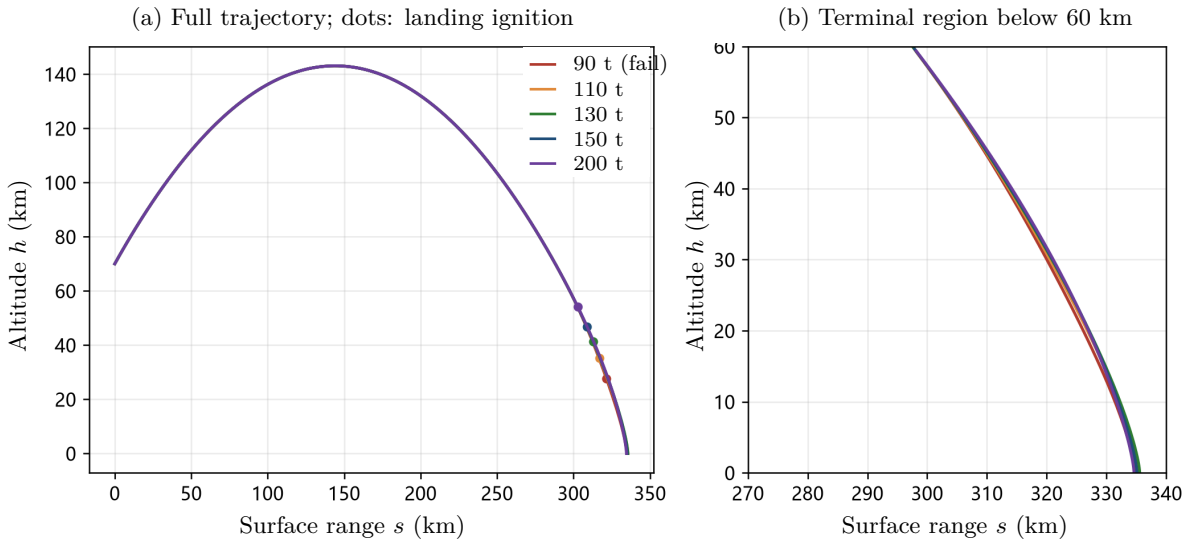


Figure 3. Baseline altitude–range trajectories. (a) Full flight histories; dots indicate terminal ignition. (b) The region below 60 km.

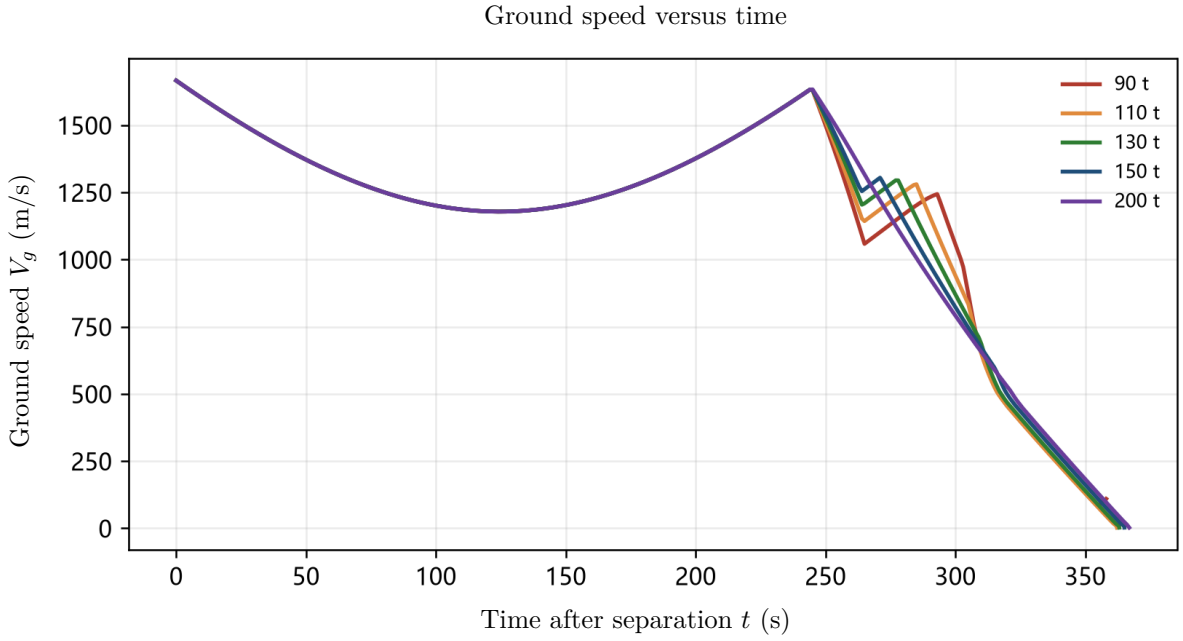


Figure 4. Ground-speed histories across the baseline propellant-load scan. Acceleration after the initial speed minimum reflects the descending arc. Low-load cases exhaust propellant before completing terminal braking.

Table 4. Reported baseline load scan; fuel entries include both oxygen and methane.

Load (t)	Outcome	V_{td} (m/s)	Entry (t)	Landing (t)	$q_{\max}$ (kPa)	$n_{\max}$	Time (s)
90	Exhausted	109.3	43.8	46.2	49.7	1.84	358
100	Exhausted	30.1	43.2	56.8	48.1	1.77	361
110	Pass	2.4	42.7	64.5	47.6	1.70	362
130	Pass	2.5	41.9	77.1	47.6	1.60	363
150	Pass	2.5	41.2	90.1	47.6	1.52	365
200	Pass	2.5	40.1	120.6	47.5	1.32	367

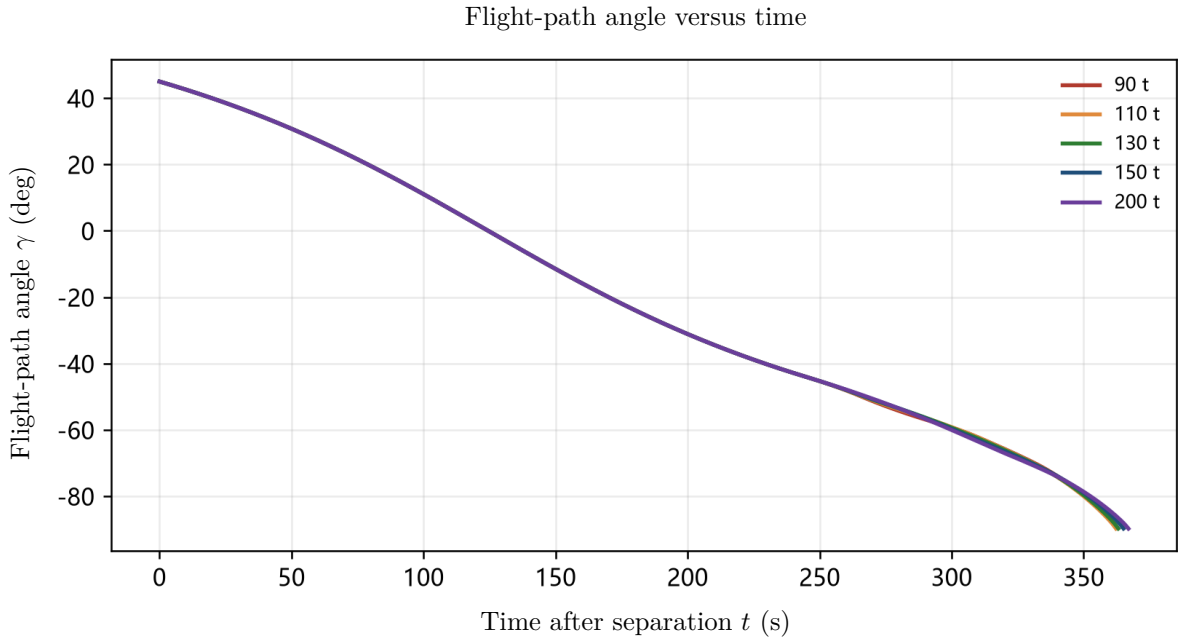


Figure 5. Flight-path-angle histories, beginning at +45° and approaching a downward vertical direction. Flight-path angle is not a separately integrated body-attitude history.

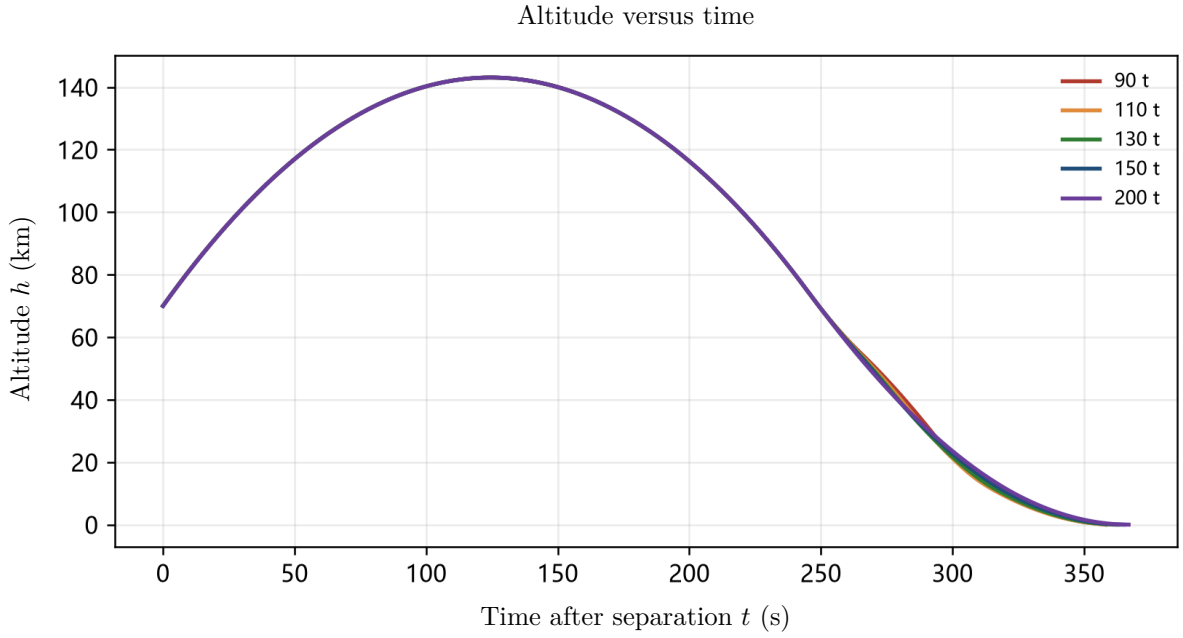


Figure 6. Altitude histories. The high-altitude ascent and return are followed by the entry pulse and extended terminal braking. A nearly straight segment of this plot is not, by itself, evidence of exactly constant speed or constant acceleration.

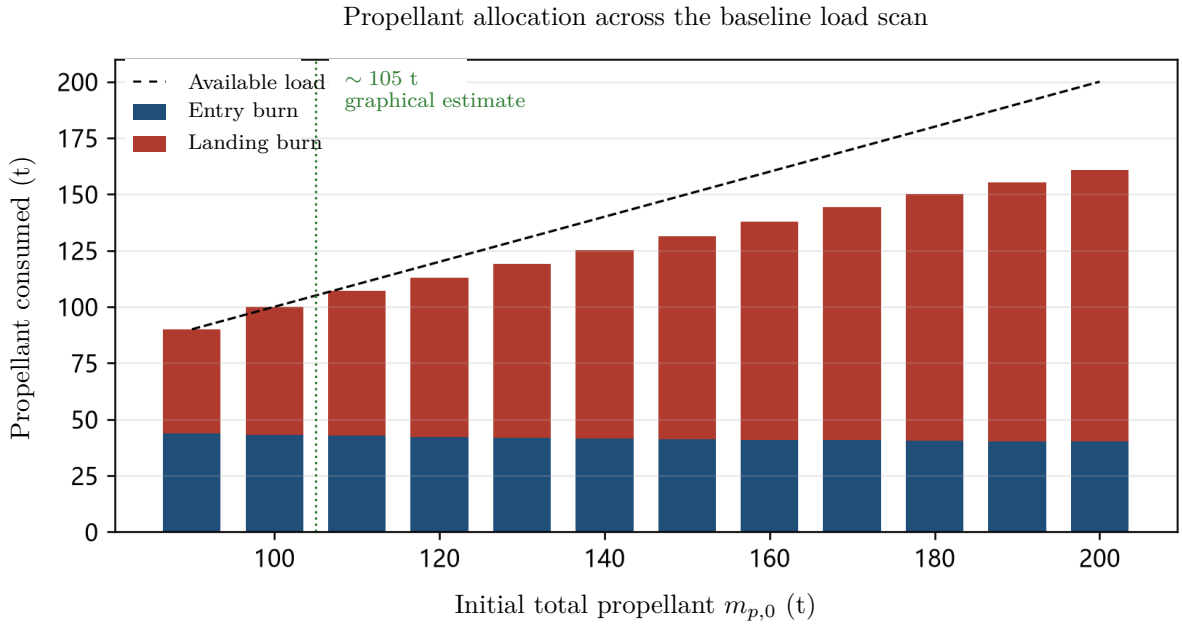


Figure 7. Baseline propellant allocation. The graphical crossing estimate is approximately 105 t, while the 1 t fine scan gives a lowest tested passing load of 104 t. Neither graphical interpolation nor a finite scan establishes a global minimum.

baseline load, rather than an analytical lower bound. The observed trend that larger loads consume more fuel applies to the reported policy and tests; it is not a theorem that every possible guidance law must have the same monotonic behavior.

4.3 Representative 150 t descent and two burns

Table 5. Simulation stage history for the 150 t baseline case.

Phase	Time (s)	Altitude (km)	Speed (m/s)	γ (deg)	Description
Ascent	0–127	70–143	1667–1165	45–0	Unpowered arc
Descent	127–245	143–75	1165–1632	0 to -44	Gravity accelerates
Entry burn	245–265	75–55	1632–1300	-44 to -50	7 MN; 41.2 t
Transition	265–272	55–47	1300–1289	-50 to -51	Unpowered, 7 s
Terminal burn	272–365	47–0	1289–2.5	-51 to -90	Throttle control; 90.1 t
Touchdown	365	0	2.5	-90	Reported cutoff event

For the model's 150 t example, entry ignition starts near 75 km, the first pulse ends near 55 km, and the unpowered transition lasts only about seven seconds. Terminal braking begins near 47 km at approximately 1,289 m/s. The model uses an effective acceleration estimate near 19.7 m/s^2 to explain a braking-distance scale of about 42 km. This is an extended stratospheric braking phase, not merely a last-second landing pulse.

Entry thrust is opposite relative airflow; terminal thrust is referenced to ground velocity. As the trajectory becomes increasingly vertical, the required thrust direction approaches upward vertical. In the ideal-attitude baseline, body alignment is prescribed rather than obtained from a full rotational simulation. Exhaust direction, thrust direction, flight-path angle, and body axis must therefore not be conflated.

Table 6. Reported entry-burn characteristics.

Quantity	Simulation value
Thrust	7 engines at 1 MN: 7 MN combined
Direction	Opposite air-relative velocity; reported vertical-reference deflection $-45.9°$ at $\gamma = -44°$
Duration	About 20 s
Propellant	40–44 t across the baseline load scan
Velocity reduction	Approximately 300–350 m/s

Table 7. Reported terminal-burn characteristics of the baseline.

Quantity	Simulation value
Thrust	5.9 MN at ignition; 7 MN peak; 2.1 MN near touchdown
Direction	Opposite ground velocity; reported vertical-reference deflection changes from $-38.7°$ to $-0.3°$
Duration	About 90 s for the 150 t baseline case
Propellant	51–121 t across the baseline load scan
Effect	About 1289 m/s to 2.5 m/s; peak pressure about 47.6 kPa

5 Wind, Attitude, and Jet-Interaction Studies

5.1 Baseline disturbance tests

The analysis distinguishes a nominal threshold from the load needed to pass its chosen perturbation set. In the baseline, 104 t has almost no residual margin. The 115 t case passes the $C_D - 15\%$ test but fails $C_D - 20\%$, while the 120 t case passes the listed baseline tests.

Table 8. Simulation baseline perturbation tests. Failed numerical entries are terminal speeds in m/s; parenthesized successful entries are remaining propellant in tonnes.

Load (t)	Nominal	$C_D - 15\%$	$C_D - 20\%$	$I_{sp} - 2\%$	Wind $\times 3$	Combined
104 t	Pass	Fail: 69.6	Fail: 92.9	Fail: 22.1	Pass	Fail: 92.5
110 t	Pass	Fail: 22.6	Fail: 45.3	Pass	Pass	Fail: 47.2
115 t	Pass	Pass: (1.1 t)	Fail: 8.2	Pass	Pass	Fail: 10.9
120 t	Pass	Pass: (3.1 t)	Pass: (1.8 t)	Pass: (5.6 t)	Pass: (6.7 t)	Pass: (1.6 t)

The combined baseline perturbation uses $C_D - 15\%$, $I_{sp} - 2\%$, and twice the reference wind. These are a finite selection of tests, not a stochastic uncertainty model. Accordingly, the model's disturbance-tested 120 t recommendation is specific to this set and guidance version; it does not supply a probability of safe recovery.

For the 150 t wind comparison, the study reports less than about 0.5% change in fuel, but a range shift near 0.5 km between adjacent wind cases. It also reports that using air-relative rather than ground-relative terminal correction can leave an approximately 8.5 m/s horizontal arrival velocity under an 8 m/s sea wind. Small fuel sensitivity therefore does not imply negligible landing-position or lateral-speed sensitivity.

5.2 Three-dimensional crosswind extension

A separate three-dimensional position/velocity model is considered with an 8 m/s crosswind. At 120 t initial load, its reported cross-track drift is 0.35 km and the summed propellant figure is about 111.8 t.

Table 9. Simulation wind comparison for the 150 t case.

Wind	Entry (t)	Landing (t)	Range (km)	V_{td} (m/s)
No wind	41.2	90.1	334.6	2.46
Reference wind	41.2	90.1	335.1	2.51
Twice reference wind	41.2	90.3	335.6	2.51

Table 10. Simulation three-dimensional extension at 120 t initial load.

Quantity	Reported value
Downrange	342.3 km
Cross-track drift	0.35 km
Entry and landing propellant	42.4 + 69.5 = 111.8 t
Two-dimensional comparison	42.3 + 70.7 = 113.0 t; reported difference 1.1 t

Rounded component and total values are transcribed as printed; their arithmetic is not silently reconciled.

The simple estimate 8 m/s multiplied by several hundred seconds overstates the reported drift because the vehicle does not instantaneously acquire the wind speed throughout the entire flight. Air density, relative-velocity geometry, and the duration of appreciable aerodynamic force matter. Terminal lateral correction also changes the result. For an order-of-magnitude check, the study uses $a_y \sim (D/m)(w_y/V)$, with an illustrative lateral acceleration near 0.5 m/s^2 acting primarily during the denser portion of the descent. The documented single profile and case do not establish a general crosswind bound.

5.3 Reaction-control and attitude estimates

A separate one-degree-of-freedom rigid-body analysis estimates rotation under applied RCS torque. The inertia surrogate is the cylinder expression $I = m(L^2 + 3R_b^2)/12$. The model uses a 0.12 MN m RCS moment and reports a 180° reorientation in about 96 s, with peak rate about 3.7°/s and cold-gas consumption 605 kg.

Table 11. Simulation one-degree-of-freedom attitude and control-authority estimates.

Quantity	Value	Interpretation
Transverse inertia	8.88×10^7 kg m^2	250 t cylinder surrogate
RCS moment	0.12 MN m	4 kN at a 30 m lever
Angular acceleration	1.35×10^{-3} rad/s^2	M/I
Half-turn duration	96 s	Bang-bang 180° maneuver
Cold-gas use	605 kg	$I_{sp} = 65$ s
Direction slew	≤ 0.63°/s	Simulation reported requirement
Aerodynamic moment	0.83 MN m at $\alpha = 5$°	$C_{m\alpha} = 0.08$/rad
TVC moment	18.3 MN m at $\delta = 5$°	Reported scale ratio about 22

The analysis additionally gives a grid-fin moment estimate of 22.8 MN m and compares its reported direction-slew requirement of 0.63°/s with a 5°/s reference limit.

The ratio of the reported 18.3 MN m TVC scale to the 0.83 MN m aerodynamic scale is about 22. Such a magnitude comparison can motivate further modeling, but does not by itself establish simultaneous force and torque feasibility, bandwidth, rate limits, or closed-loop stability. The model's direction-slew figure and RCS maneuver must likewise not be substituted for a fully coupled rotational history.

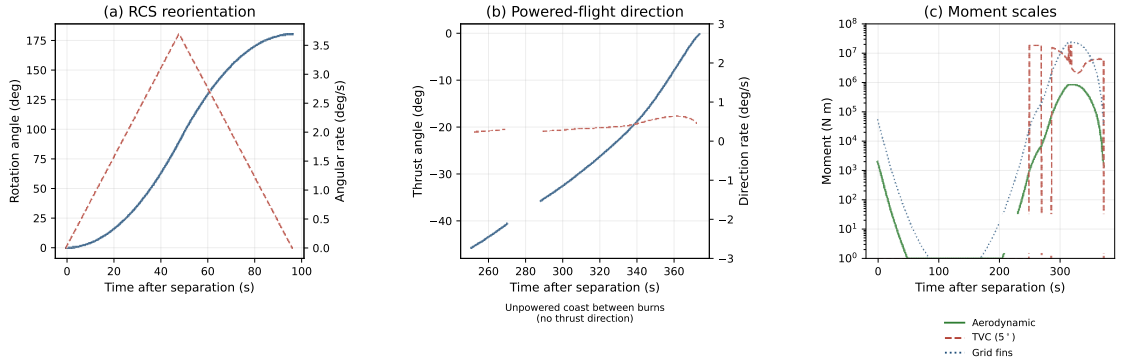


Figure 8. Attitude and moment-scale estimates. (a) One-degree-of-freedom RCS reorientation. (b) Required thrust direction during powered flight; the gap corresponds to the unpowered coast. (c) Aerodynamic, TVC, and estimated grid-fin moment scales. The traces are reproduced from plot pixels without reconstructing obscured portions. These separate estimates do not constitute a fully coupled attitude solution.

5.4 Jet-interaction parameterization

The model uses a thrust-coefficient-based correction,

$$C_T = \frac{T}{qA}, \qquad C_D^{eff} = C_D(M)\left[1 + k_{jet}\frac{C_T}{1 + C_T}\right]. \tag{10}$$

The tested k_{jet} values are 0, 0.3, and 0.6. Retropropulsion and grid-fin literature provide the physical context for considering such effects [5, 6, 10], but the chosen coefficients are scenario parameters rather than a calibrated database for this booster.

Table 12. Simulation jet-interaction sensitivity at 120 t initial load.

Jet coefficient	Entry (t)	Landing (t)	Total (t)
0	42.4	70.5	112.8
0.3	42.4	69.5	111.8
0.6	42.4	69.6	111.9

The jet coefficient is a parametric assumption, not a CFD-calibrated property of the booster.

Table 13. Summary of separate model extensions.

Model	Propellant at 120 t	Touchdown	Qualification
2D reference	113.0 t	2.5 m/s	Ideal attitude, no jet correction
3D with crosswind	111.8 t	2.5 m/s	0.35 km cross-track drift
3D jet-parameter cases	111.8–112.8 t	2.5 m/s	About 1 t variation
Simulation load recommendation	112 t	2.5 m/s	120 t includes tested disturbance allowance

The calculations indicate approximately 1 t variation across the listed jet cases. The analysis does not calculate bottom heating, surface pressure distribution, or reusable thermal protection. The limited fuel difference must not be interpreted as a comparably small thermal or structural effect.

6 Parameterized Guidance Optimization

6.1 Search method and decision variables

The four-parameter direct shooting search uses SciPy's derivative-free Powell method and grid refinement. CasADi/IPOPT is not used for these results.

Table 14. Four variables in the reported Powell and grid search.

Variable	Meaning	Range
h_e	Entry-ignition height	35–82 km
v_{cut}	Entry cutoff speed	450–1150 m/s
k	Landing-ignition height multiplier	0.95–1.60
V_{td}	Target touchdown speed	0.5–3.0 m/s

The parameters are entry ignition height h_e, entry cutoff speed v_{cut}, the terminal-ignition multiplier k, and target speed V_{td}. Each trial corresponds to a simulated trajectory with those rules. The model describes the penalty objective as

$$J = \dot{m}\, t_{burn} + 50 \max(0, V_{touch} - 2.5) + 2 \max(0, q_{max} - 50 \text{ kPa}). \tag{11}$$

The objective requires consistent unit normalization and calibrated penalty weights before independent reproduction. This is a parameterized guidance search with penalties, not a demonstrated solution of the unrestricted continuous optimal-control problem.

The reported constraints are touchdown speed below 2.5 m/s, peak dynamic pressure at most 50 kPa, and no consumption beyond the carried load. A run that exhausts propellant before meeting the terminal requirement is counted as a failure.

6.2 Late and shorter entry pulse

Table 15. Simulation comparison between baseline and parameterized guidance.

Item	Baseline guidance	Parameterized search
Entry burn	75 to 55 km; about 42 t	55 to 40 km; about 27 t
Terminal braking	From about 47 km; 70–121 t	Pressure-triggered; 69–87 t
Lowest reported feasible load	104 t	96 t
Consumption at 120 t	111.8 t	108.5 t
Peak pressure	47.6 kPa	47.6 kPa

The reported parameterized solution moves entry ignition downward from 75 km to about 55 km and ends the pulse near 40 km. Entry propellant decreases from roughly 42 t to 27 t in that comparison. The analysis attributes the saving to avoiding an extended high-altitude burn where aerodynamic braking is weak and gravity continues to act during thrusting.

This explanation is specific to the guidance strategy and its constraints. Delaying ignition too far can also remove the remaining stopping distance or increase the pressure peak. Reusable-launcher guidance literature motivates the interaction of timing, control, and terminal requirements [7, 8, 9]; it does not turn the finite parameter search into a global optimum proof.

6.3 Reported optimized disturbance allowance

Under the parameterized guidance, the study reports 96 t as the lowest nominal successful load in its test, while 110 t passes the selected optimized perturbations. At 110 t, reported

Table 16. Simulation perturbation results for parameterized guidance. Numeric successful entries are consumed propellant in tonnes.

Load (t)	Nominal	$C_D - 15\%$	$C_D - 20\%$	$I_{sp} - 2\%$	Combined
96 t	Pass: 95.9	Fail	Fail	Fail	Fail
104 t	Pass: 100.2	Fail	Fail	Pass	Fail
110 t	Pass: 103.3	Pass: 107.3	Pass: 108.7	Pass: 104.6	Pass: 108.8
112 t	Pass: 104.4	Pass: 108.3	Pass: 109.7	Pass: 105.7	Pass: 109.8

consumed propellant is 103.3 t nominally and 108.8 t in the combined case. At 112 t these numbers are 104.4 t and 109.8 t. The numerical entries represent consumed propellant, whereas parenthesized successful values in the baseline perturbation table represent residual mass; their units and meanings must be kept distinct.

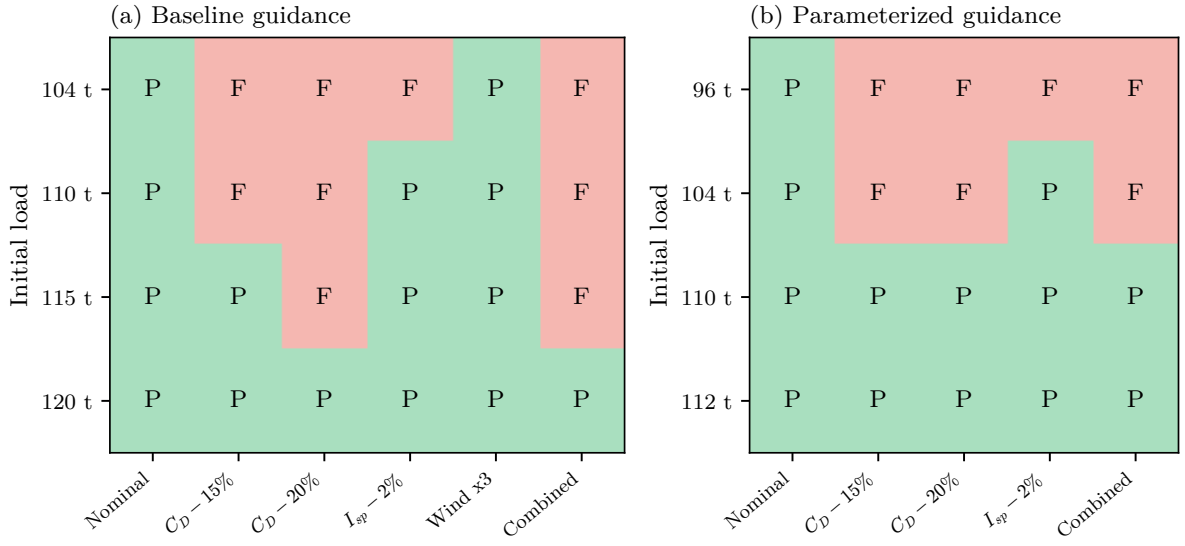


Figure 9. Pass/fail patterns for the tested cases: (a) baseline guidance and (b) parameterized guidance. P and F denote the model's reported outcomes. These are categorical views of the tabulated results, not new simulation runs or probability estimates.

7 Area Coupling, Engine Response, and Staging

7.1 Attitude-dependent effective area

The final model refinement replaces a fixed reference area with

$$A_{eff} = A\cos^2\alpha + (Ld)\sin^2\alpha, \qquad A = 38.48\ \text{m}^2, \quad Ld = 455\ \text{m}^2. \tag{12}$$

A first-order attitude lag with a reference time constant of 1 s allows a mismatch between the prescribed alignment and body direction. Without any mismatch, an ideal alignment model could hide the influence of the area law. A separate comparison tests a 3 s lag in its later comparison.

The results indicate fuel reductions of about 1.3–1.4 t in these particular area-law cases. Increased projected area can provide additional drag, but this is not a general claim that attitude error is beneficial: loads, torques, heating, and control feasibility can also change. The analysis does not provide a complete validated aerodynamic force-and-moment model for those effects.

Table 17. Simulation area-law comparison before the final combined model.

Load (t)	Without area law (t)	With area law (t)	Difference (t)	$q_{\max}$ (kPa)
110	109.2	107.8	-1.4	45.6
120	114.8	113.5	-1.3	44.7

7.2 Design and acceptance pressure thresholds

The later model uses a 47 kPa guidance trigger and a 50 kPa acceptance screen. Its listed pressure peaks lie around 44.7–45.6 kPa, leaving roughly 5 kPa below the acceptance value. This is an assigned design gap, not a quantified confidence interval for drag and wind uncertainty.

7.3 Finite response and the throttle-floor problem

The refined model adds a 0.5 s ignition delay, a 0.1 s first-order response, and a throttle slew limit of 1 s^{-1}. In its first attempt at this refinement, 110 t and 120 t cases exhausted propellant and reached reported terminal speeds of roughly 160–180 m/s.

The subsequent diagnosis identifies the throttle floor rather than delay alone as a central difficulty. The terminal law requested only about 10–18% of the seven-engine nominal thrust, while seven simultaneously enabled engines at 30% supply 2.1 MN. The model describes excessive braking, a low-speed condition near 2.6 km, renewed descent, and eventual exhaustion.

The reported remedy is staged engine selection: seven engines during entry, three in the intermediate part, and one in the late terminal portion. Each enabled engine retains the stated 30–110% throttle range. This reduces absolute thrust increments and permits finer terminal modulation. It is a change in the control arrangement, not evidence that a single engine can hover against the dry weight; a short transient arrival and a stable supported landing are different questions.

7.4 Combined results and final propellant recommendation

Table 18. Simulation ablation of area law, finite engine response, and engine-count staging.

Model	Fuel at 110 t	Fuel at 120 t	V_{td} (m/s)	$q_{\max}$ (kPa)
Ideal v3 reference	109.2 t	114.8 t	2.40	45.6
Area law added	107.8 t	113.5 t	2.46	45.6
Engine response and staging	109.0 t	114.7 t	1.96	44.9
Both effects: combined v4	106.6 t	112.7 t	2.02	44.9
Combined, attitude lag $\tau = 3$ s	107.9 t	113.7 t	2.01	44.7

For the 110 t combined case, entry and terminal propellant are tabulated as 45.2 t and 61.5 t, alongside a total of 106.6 t. The 0.1 t discrepancy between the displayed sum and total should be resolved using unrounded records.

The final combined v4 row reports 106.6 t consumption at 110 t initial load and 112.7 t at 120 t, with a 2.02 m/s touchdown entry and 44.9 kPa pressure peak in the summary. The separate 3 s attitude-lag variant consumes more than the 1 s reference. These refinements should be compared within their documented versions, rather than mixing individual component numbers from unrelated cases.

The final load scan records 96 t and 100 t as unsuccessful, followed by 102 t arriving at 2.02 m/s after consuming 101.7 t. Its recommendation is 110–112 t, retaining an allowance above the lowest reported passing case. The earlier 96 t value belongs to the more idealized parameterized model, while the earlier 104 t value belongs to baseline guidance.

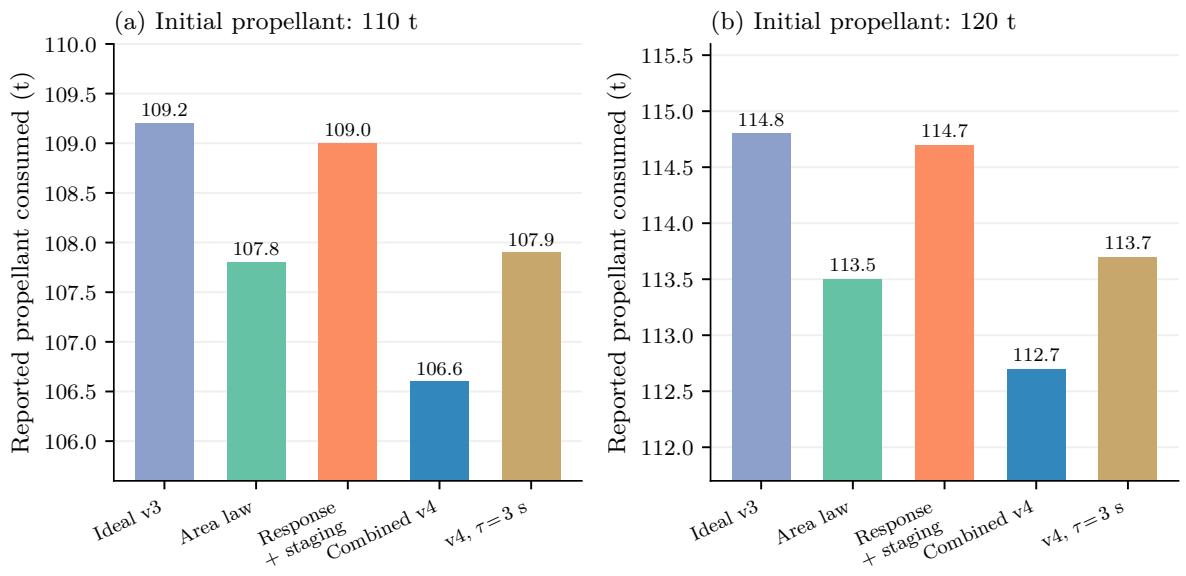


Figure 10. Fuel consumption for model variants at (a) 110 t and (b) 120 t initial propellant. Values are plotted directly from the ablation table. Adding engine response also changes engine-count staging, so the difference is not an isolated physical effect of response delay alone.

Table 19. Simulation final load scan with area law, finite response, and staged engine selection.

Load (t)	Outcome	Consumed (t)	V_{td} (m/s)
96 t	Fail	96.0	37.5 m/s
100 t	Fail	100.0	11.3 m/s
102 t	Pass	101.7	2.02 m/s
110 t	Pass	106.6	2.02 m/s
120 t	Pass	112.7	2.00 m/s

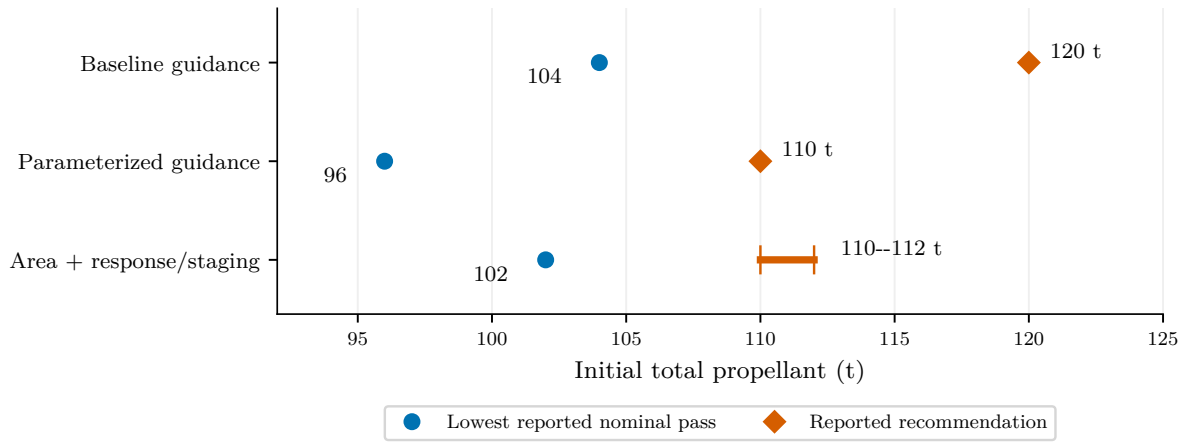


Figure 11. Version-specific load thresholds and recommendations reported in the study. The circles are the lowest reported nominal passes; the recommendations refer to their respective reported test sets or design allowances. Different model and acceptance settings prevent interpreting this as a universal ranking of minimum fuel.

8 Discussion: Strengths and Limitations

8.1 Physical interpretation of the study trends

Three recurring effects organize the results. First, entry and terminal burns play different roles: the baseline entry pulse removes a modest part of the speed, while terminal braking must close the remaining velocity and altitude budget. Second, additional propellant changes the mass being decelerated, so a heavier initial load can require earlier braking and consume much of the added mass. Third, state-dependent guidance must be compatible with absolute thrust floors, finite response, and the number of active engines.

The model's wind comparison also separates fuel efficiency from landing accuracy. Similar consumed mass does not imply similar lateral speed or range. Its area and jet investigations show why a change in aerodynamic representation can alter the apparent fuel requirement even without changing the basic dimensions or propellant species.

8.2 Strengths of the modeling sequence

The study compares several physically motivated refinements instead of presenting one unexplained fuel number. It distinguishes air-relative and ground-relative velocities, shows representative stage histories, includes finite perturbation sets, and diagnoses a throttle-floor problem that an ideal continuous-thrust law can hide. Keeping separate tables for baseline guidance, parameterized guidance, and actuator refinements makes those comparisons interpretable.

The separate RCS and moment estimates provide useful scale checks, while the three-dimensional example highlights cross-track motion absent from the simplest plane model. These are useful components of further analysis, even though they do not collectively demonstrate a complete closed-loop recovery simulation.

8.3 Limitations and evidence still required

The numerical results have not been independently rerun. Complete executable programs, $C_D(M)$ anchor values, high-altitude extension details, and penalty normalization are required for independent numerical reproduction.

The displayed velocity equations and angular definitions require implementation review, as described in equations (4) and (8). Peak load factor is reported in the load scan, but a full stress distribution, structural capacity, and thermal-protection analysis are not available. No temperature history is manufactured from the pressure result.

The early ideal-engine and ideal-alignment assumptions do not apply to every late model. Conversely, the later throttle range, delay, and lag do not establish engine-out behavior or actual flight hardware capability. The model's one-dimensional attitude estimates are post hoc, and the final first-order lag is not a complete coupled rigid-body controller. Deployment, actual foot velocities, deck motion, leg compression, friction, and post-contact stability remain outside the demonstrated result.

Finally, finite successful perturbation sets justify only their tested cases. They do not establish a failure probability, a confidence interval, or a universal robustness margin. The combined refined model's 110–112 t recommendation should remain a reported engineering-study recommendation until the same physical model, guidance, and uncertainties are evaluated together.

9 Conclusion

This simulation study describes the return of a seven-engine, 130 t dry-mass booster from 70 km and 6,000 km/h with an initial upward flight-path angle of 45°. Its reported ascent reaches about

143 km before the entry and terminal burns complete the descent. Baseline entry propellant remains near 40–44 t, while terminal consumption grows strongly with carried mass.

The documented conclusions depend on model version. Baseline guidance reports a lowest passing tested load of 104 t and a disturbance-tested recommendation of 120 t. Parameterized Powell/grid guidance reports 96 t nominally and recommends 110 t for its chosen tests. The later combined area-law, finite-response, and staged-engine model reports a 102 t nominal passing case and recommends 110–112 t. At 110 t its summary gives 106.6 t consumed and 2.02 m/s touchdown speed.

The results highlight the interaction of ignition timing, mass, drag, throttle floor, and engine selection. They remain empirical results of the tested models rather than mathematical global minima or a demonstration of complete shipboard landing safety.

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Report on the Use of AI

AI assistance was used for language editing, equation and table typesetting, figure-label formatting, and LaTeX preparation. No independent trajectory recomputation is attributed to this assistance. The team must verify the numerical results and references, insert its assigned team number, and complete the disclosure record required by the competition.